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Computational Estimates of Fluoride Affinity of Boron-Based Anion Receptors

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Ab initio calculations were carried out to study the structure–property relationship of anion receptors to establish a guidance for developing advanced electrolyte for lithium-ion batteries. A group of 25 well-selected anion receptors was investigated to illustrate the impact of structural rigidity and the substitution of electron-withdrawing groups, such as F and CF_3, as well as the substitution of alkoxyl groups. The calculation results showed that the fluoride affinity of an anion receptor can be reduced by the introduction of alkoxyl groups to the boron center and that the fluoride affinity increases with the substitution of electron-withdrawing groups.
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Among the available battery technologies, lithium-ion batteries have the highest energy density and hence have excellent potential for applications in stationary energy backup systems[1,2] and hybrid electric vehicles (HEVs).[3-5] However, the commercial application of lithium-ion batteries in HEV has not yet occurred due to their insufficient life, potential fire hazard, and high cost. Hence, there is a major research and development effort in developing low cost lithium-ion cell chemistry, including low cost materials and cell components with improved power capabilities. A major cost reduction of lithium-ion batteries for HEV can be practically achieved by improving the power density of lithium-ion cells so that the size of the battery pack can be reduced while still meeting the power requirements for transportation applications. Besides the high rate positive and negative electrode materials, a properly formulated electrolyte can also greatly enhance the power capabilities of lithium-ion cells.[6-8]

Anion receptors (ARs) can specifically complex with anions to reduce the attraction between Li^+ and the anions and enhance the ionic conductivity of electrolytes. The ARs that have been investigated in lithium-ion cells are boron-based (B-based) compounds that include boranes, borates, borinates, and boroles. In the B-based ARs, the B atom generally adopts an sp^2 hybrid configuration, forming three σ bonds connected to alkyl or alkoxyl groups and one empty π orbital perpendicular to the σ plane. The empty π orbital allows the molecule to accept an electron pair donated by other molecules or anions and to form a π-π bond between the AR and the electron donor. In nonaqueous electrolytes for lithium-ion batteries, the ARs can complex with anions, such as F^-, PF_6^-, and BF_4^-, to reduce the attractive interaction between Li^+ and the anions. Hence, an improvement in the ionic conductivity can be reasonably expected with the addition of various ARs to the electrolytes. Moreover, ARs were also reported to improve the thermal stability of the lithiated graphite[9-11] and the cycling performance of lithium-ion cells.[8,12,13] There are also reports that ARs can be used as the precursors to develop lithium salts for applications in the electrochemical devices.[14-16]

The ionic conductivity of commonly used $LiPF_6$/carbonate electrolytes is quite high (>6 mS/cm at 25°C), and the bottleneck of the power capabilities of lithium-ion cells lies on the diffusion of lithium or lithium ion in the bulk electrode materials and through the solid electrolyte interphase (SEI) but not through the liquid electrolyte phase. The power capabilities of lithium-ion cells can be improved by adding a controlled amount of tris(pentafluorophenyl)borane (TPFPB) to the electrolyte.[8] The idea is to add optimized amounts of TPFPB that is just enough to dissolve the insulator, LiF, out of the SEI layer, as shown in the following equation

$$\text{TPFPB} + \text{LiF} \rightarrow \text{TPFPB} - \text{F}^- + \text{Li}^+ \quad [1]$$

When more TPFPB was added, the excess TPFPB would react with $LiPF_6$ and generate PF_5 as follows

$$\text{TPFPB} + \text{PF}_6^- \rightarrow \text{TPFPB} - \text{F}^- + \text{PF}_5 \quad [2]$$

Sloop et al. reported that PF_5 is a catalyst for accelerating the ring-open polymerization of ethylene carbonate.[17] Hence, the excess amount of TPFPB indirectly caused the further thickening of the SEI layer, and the power capability of the cell greatly suffered.[8] Therefore, the concentration of TPFPB needs to be optimized case by case according to the specific surface area of the materials and the loading ratio of the materials to the electrolyte.

For the effort reported here, we proposed to develop alternative or weaker ARs that can dissolve LiF while substantially suppressing the decomposition of PF_6^- with excess ARs. To obtain the broadest possible range of B-based compound configurations, including those that have not yet been commercially synthesized, ab initio calculations were carried out to estimate the fluoride affinities (FAs) of ARs, as well as the impact of the substitution groups on the FAs.

Experimental

All the calculations reported in this work were carried out using the Gaussian03 computational package.[18] The full geometry optimization and the calculation of the zero-point energy correction and the thermal correction to the standard free energy at 298 K were performed using the Hartree–Fock theory with a basis set of 6-31G(d,p) [HF/6-31G(d)]. The single-point electric energy was determined from ab initio calculations using B3LYP, a well-known hybrid model of density functional theory with the Becke three-parameter function for electron exchange[19] and the Lee–Yang–Parr function for electron correlation.[20] The basis set for single-point energy calculations was 6-31 + G(d). The impact of the nonaqueous solvents on the calculation results was also studied using the polarization continuum model.

The electric energy and the standard free energy at 298 K can be approximately expressed as a sum of four terms, as shown in Eq. 3-5

$$E = E_{\text{elect}} + \text{ZPE} + E_{\text{vrt}} \quad [3]$$

$$H = E + PV = E + RT \quad [4]$$

$$G = H - TS \quad [5]$$

In the above equations, E is the electric energy, H is the enthalpy, G is the standard free energy, E_{elect} is the single-point electric energy calculated at B3LYP/6-31 + G(d), ZPE is the zero-point energy correction calculated at HF/6-31G(d), E_{vrt} is the thermal correction of vibration, rotation, and translation energies, RT comes from the ideal gas equation, $PV = nRT$ (with $n = 1$), and TS is the contribution of entropy to the standard free energy and is calculated at HF/6-31G(d).

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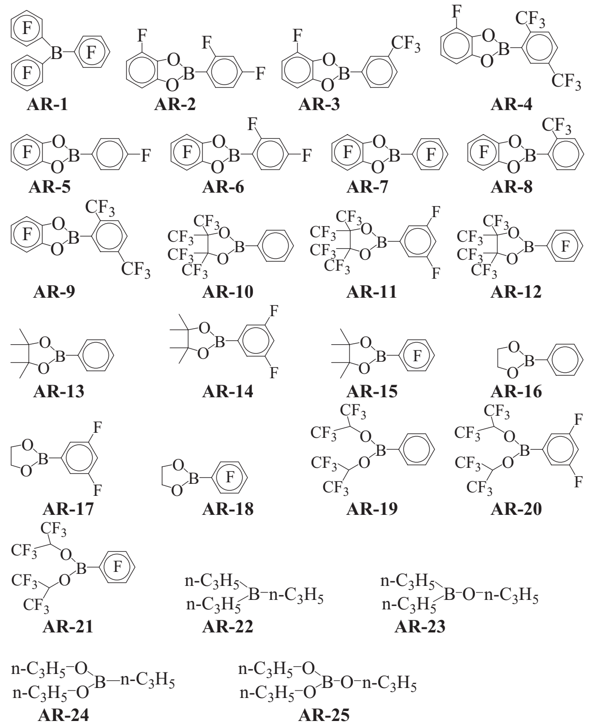

Figure 1. Molecular structures of ARs investigated.

The FA of an AR is defined as the change in electric energy when the AR combines with a F^- anion, as shown in Eq. 6

$$AR + F^- \rightarrow AR - F^- \quad [6]$$

$$FA = E_{AR} + E_F - E_{AR\text{-}F} \quad [7]$$

$$\Delta G^{298\ K} = G_{AR\text{-}F} - G_{AR} - G_F \quad [8]$$

In Eq. 7 and 8, FA is the defined FA of the given AR; $E_{AR\text{-}F}$, E_{AR}, and E_F are the calculated electric energies of AR-F^-, AR, and F^-, respectively; $\Delta G^{298\ K}$ is the change in the standard free energy at 298 K; $G_{AR\text{-}F}$, G_{AR}, and G_F are the calculated standard free energies of AR-F^-, AR, and F^-, respectively.

Results and Discussion

Figure 1 shows the molecular structures of the ARs explored in this work, most of which have aromatic substitution groups and have been previously reported by Lee et al.[21] The variables covered in this major group include the degree of fluorination on the aryl group, the rigidity of the substitution groups, and the fluorination of the substituted alkoxyl groups. The other anion receptors from AR-22 to AR-25 were selected to illustrate the impact of alkoxyl groups vs alkyl groups.

To investigate the impact of solvation on the calculated results, TPFPB was used as an example, and the change in free energy was calculated using solvents with different dielectric constants. Figure 2 shows the calculated free energy change $[\Delta G(\text{sol})]$ of the ARs when complex with F^- in different solvents as a function of the dielectric constant of the solvent. Figure 2 shows a strong correlation between the calculated $\Delta G(\text{sol})$ and the dielectric constant of the solvent. The dramatic impact of the solvent was observed for solvents with small dielectric constants (<20). When the dielectric constant was larger than 20, the impact of the solvent was stabilized and was not as significant as those solvents with low dielectric constants. The dielectric constant of actually used electrolytes depends on the composition of the solvents and is somewhat close to acetonitrile. Therefore, acetonitrile was selected as the model solvent to investigate the impact of solvation on different ARs.

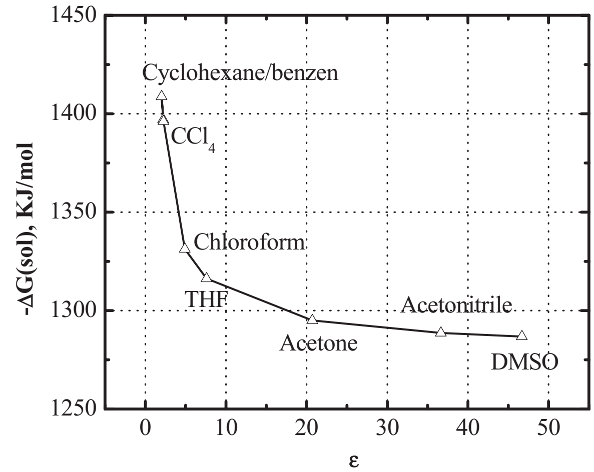

Figure 2. Correlation between the calculated free energy change in solvent to the dielectric constant of the solvent investigated.

Table I shows the calculated FAs and the change in free energies at the gas phase ($\Delta G^{298\ K}$) and the change in free energies in acetonitrile $[\Delta G(\text{sol})]$ when the AR received an F^- anion and formed the corresponding complex. FAs are mostly positive and $\Delta G^{298\ K}$s are mostly negative, although they are all indicators of the capability to accept a F^- anion. This difference originates from the definition, as shown in Eq. 7 and 8. A better AR for the F^- anion generally has a greater positive FA value and a more negative value for $\Delta G^{298\ K}$. Figure 3 shows (a) plot of $\Delta G(\text{sol})$ vs $\Delta G^{298\ K}$ and (b) plot of –FA vs $\Delta G^{298\ K}$. It is clear that the calculated FA agreed well with the $\Delta G^{298\ K}$. For the ARs examined in this work, both FA and $\Delta G^{298\ K}$ can be used as a quantitative index for the capability of accepting a F^- anion. However, the change in the free energy was chosen for the

Table I. Calculated fluorine affinities and the change in standard free energy for different ARs.

Molecular	FA (kJ/mol)	$\Delta G^{298\ K}$ (kJ/mol)	$\Delta G(\text{sol})$ (kJ/mol)
AR-1	406.0	−368.4	−1288.7
AR-2	297.9	−264.8	−1233.5
AR-3	300.34	−280.5	−1242.3
AR-4	320.7	−294.5	−1247.9
AR-5	327.1	−297.1	−1249.1
AR-6	338.7	−305.9	−1255.0
AR-7	370.5	−338.0	−1272.8
AR-8	335.9	−310.4	−1259.1
AR-9	363.9	−337.5	−1269.7
AR-10	324.2	−289.8	−1246.6
AR-11	355.7	−322.6	−1267.5
AR-12	384.3	−348.2	−1286.1
AR-13	199.2	−170.4	−1159.8
AR-14	226.4	−193.5	−1168.4
AR-15	253.5	−218.2	−1185.2
AR-16	195.2	−166.0	−1169.4
AR-17	227.8	−198.2	−1185.1
AR-18	255.4	−230.6	−1207.1
AR-19	353.7	−321.5	−1267.6
AR-20	382.5	−350.3	−1286.5
AR-21	411.8	−382.6	−1310.6
AR-22	227.4	−190.1	−1163.8
AR-23	196.9	−162.1	−1140.4
AR-24	185.2	−149.5	−1128.3
AR-25	−22.5	36.7	−1001.1
PF_5	369.6	−327.9	−1340.8

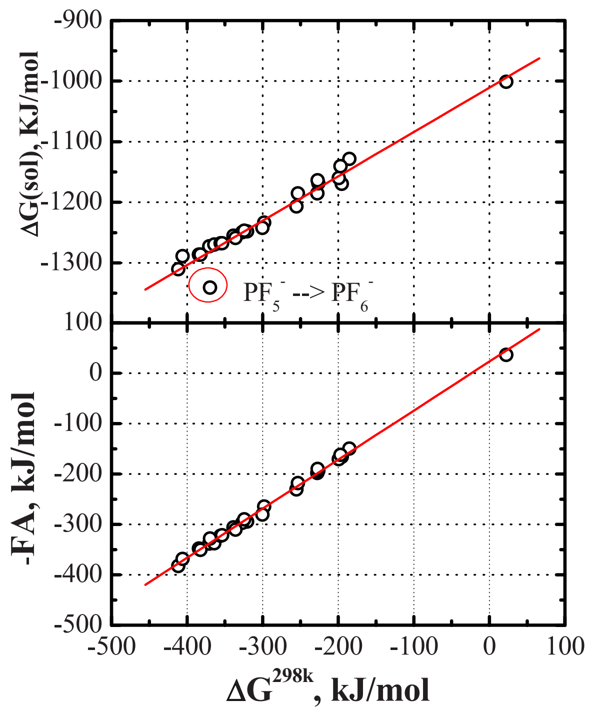

Figure 3. (Color online) Correlation of (a) the calculated free energy change in solvent $[\Delta G(\text{sol})]$ to the free energy change in gas phase ($\Delta G^{298\ \text{K}}$) and (b) the FA to the change in free energy ($\Delta G^{298\ \text{K}}$) when the AR complexes with a F^- anion.

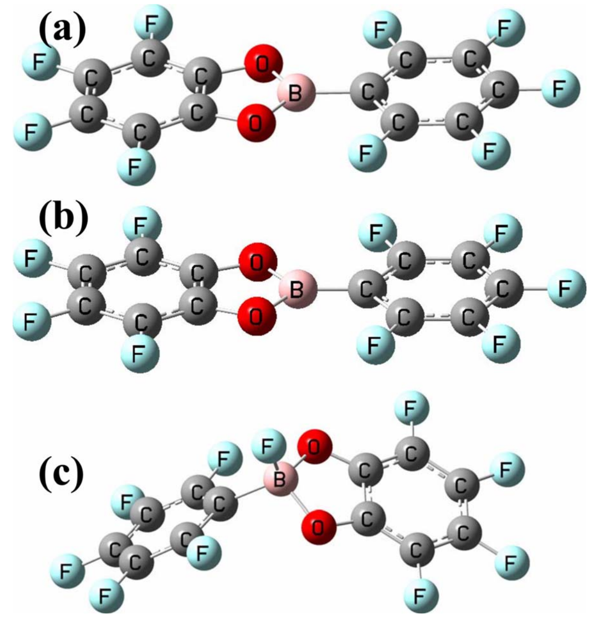

Figure 4. (Color online) Computationally optimized molecular geometry of (a) AR-7, (b) AR-7⁻ (accepting an electron), and (c) AR-7-F⁻ (accepting F⁻ anion).

following discussions because the change in the free energy (ΔG) is a better indicator for judging the likelihood or spontaneity of the reaction of interest. Figure 3a shows that the calculated free energy change in acetonitrile also agreed well with the calculated free energy change in the gas phase except for PF_5. A reasonable explanation for the case of PF_5 is that the impact of solvation tends to have a bigger impact on small or hard molecules/anions such as PF_5, PF_6^-, and F^-. It is shown that the ARs studied in this work all showed weaker FA in the nonaqueous solvents than PF_5. However, a weaker AR is still desired to reduce the possibility of PF_6^- decomposition with the presence of ARs.[8] It can also be seen that there is consistent difference of about 900 kJ/mol between $\Delta G^{298\ \text{K}}$'s and $\Delta G(\text{sol})$'s. This is mostly originated from the salvation process of F^- in the solvent (acetonitrile), and hence the difference is almost the same for different ARs.

Table I lists a wide range of ARs with $\Delta G(\text{sol})$ values ranging from -1340 to -1001 kJ/mol. Our objective was to understand the structure–property relationship of the ARs using this limited set of examples. The functionality of ARs comes from the empty orbital in the B atom, which can also serve as the acceptor for electrons. We tried to correlate the FAs to the electron affinities, but without success. This discrepancy can be explained by the difference between molecular configurations when accepting an electron or a F^- anion, as shown in Fig. 4. In the original state, the B atom in the AR adopts an sp^2 hybrid configuration with the empty π orbital perpendicular to the σ plane (see Fig. 4a). When the AR accepts an electron, the electron is directly injected into the π orbital while the B atom still maintains its sp^2 hybrid configuration (see Fig. 4b). However, the B atom has to adopt an sp^3 configuration when a F^- anion is attached to it, resulting in a tetrahedral structure (see Fig. 4c). Hence, we need to investigate the FAs explicitly, rather than electronic affinities or Lewis acid strength. This also implies that the flexibility of the substitution groups can affect the energy barrier of the configuration transition and pose a significant influence on the ability to accept F^- anions.

Figure 5 shows a direct comparison of the change in free energy $[\Delta G(\text{sol})]$ for three ARs showing the impact of their structural rigidity. The three ARs are AR-7, AR-12, and AR-21. The three ARs for comparison have a pentafluorophenyl group directly bonded to the B atom and other substitution groups attached to B via two O atoms. Among the ARs for comparison, AR-21 with two hexafluoroisopropoxyl groups bonded to B atoms has the highest flexibility and shows the highest $-\Delta G(\text{sol})$, meaning a greater ability to accept the F^- anion. With two hexafluoroisopropyl groups connected by a σ bond, AR-12 can be expected to be more rigid than AR-21, and the change in free energy for AR-12 is -1286.1 kJ/mol, smaller than that of AR-21. Figure 4 also shows that $-\Delta G(\text{sol})$ can be further reduced by replacing the substitution groups with a more rigid tetrafluorophenyl group (AR-7).

Figure 6a shows the impact of fluorination on the FAs of ARs. Data set A contains representatives for the demonstration of fluorination on the phenyl group directly bonded to the B atom; data sets B and C contain representatives for illustrating the impact of substitution of a H atom with the trifluoromethyl group; and finally data

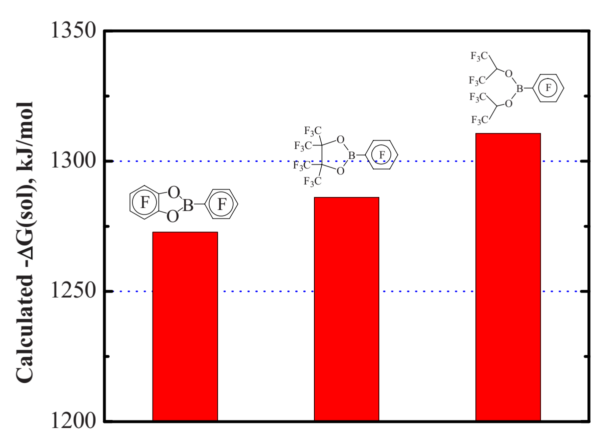

Figure 5. (Color online) A comparison on $\Delta G(\text{sol})$ for ARs showing the impact of structural rigidity.

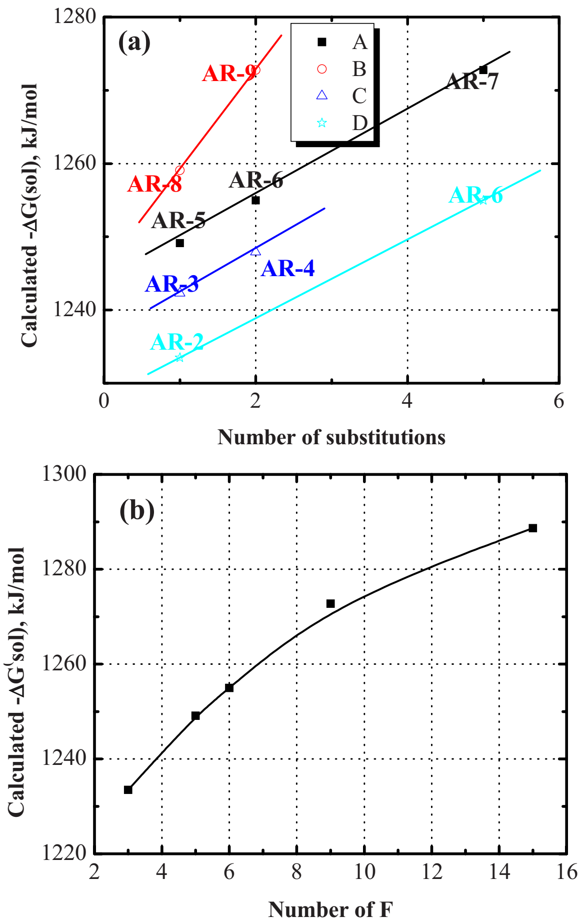

Figure 6. (Color online) (a) Impact of the substitutions on the phenyl groups on the calculated $-\Delta G(\mathrm{sol})$. Data set A contains representatives for demonstration of fluorination on the phenyl group directly bonded to the B atom; data sets B and C contain representatives for illustrating the impact of substitution of the H atom with a trifluoromethyl group; and finally data set D represents the impact of the fluorination on the phenyl group bonded to B atom via O atoms. (b) Correlation of the number of fluoride substitutions and the calculated $-\Delta G(\mathrm{sol})$.

set D represents the impact of the fluorination on the phenyl group bonded to a B atom via O. Figure 6a clearly shows that the substitution of electron-withdrawing groups such as –F and $-CF_3$ can increase the $-\Delta G(\mathrm{sol})$ of ARs. It is also shown the substitution of $-CF_3$ has more impact than that of fluoride substitution. The data sets A and D have the same slope vs the number of fluoride substitutions, indicating that the fluorination on both phenyl groups has the same impact on the FAs of ARs. This can be further confirmed in Fig. 6b, which shows the change in free energy $[-\Delta G(\mathrm{sol})]$ vs the total number of fluoride atoms substituted on both phenyl groups. Figure 6 implies that a low level of fluorination is preferred to reduce its FAs below that of PF_5. However, the redox potential of the slightly fluorinated aromatic compounds is generally lower than 4.0 V vs Li^+/Li, and hence such ARs are less practical in lithium-ion cells.[22,23]

To raise the redox potential of ARs, one can simply replace one of the phenyl groups with alkyl or alkoxy groups such as AR10–21. Figure 7 shows the change in free energy $[-\Delta G(\mathrm{sol})]$ for AR10-18. The variables shown in this set of examples are (*i*) the number of fluoride substituted to the phenyl group and (*ii*) the substitution of H with CH_3 and CF_3 for the alkoxyl groups. Consistent with Fig. 6, Fig. 7 shows that the FAs of the ARs increase with the degree of fluorination on the phenyl group. It is also shown that the $-\Delta G(\mathrm{sol})$

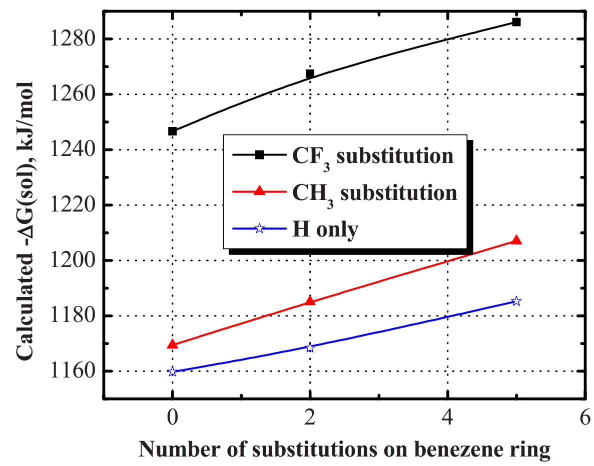

Figure 7. (Color online) Impact of the number of substitutions on the alkoxyl group on the calculated $-\Delta G(\mathrm{sol})$.

of CF_3-substituted ARs is substantially higher than that of H- and CH_3-substituted ones.

It can be concluded from the above discussion that the substitution of strong electron-withdrawing groups, such as F and CF_3, on the alkoxyl groups can also increase the FAs of ARs primarily due the decrease in electron density on B atom by the introduction of strong electron-withdrawing groups. Alternatively, the FAs can be reduced by introducing electron-abundant atoms such as O to bridge the B atom and the substituted alkyl groups, forming alkoxyl groups. The impact of the oxygen bridge on the FAs is clearly shown in Fig. 8.

Conclusion

The FAs of a group of 25 ARs were investigated using ab initio calculations, demonstrating that theoretical calculations could be a powerful tool for the development of advanced ARs for lithium-ion batteries. Several trends were consistently illustrated in this work.

1. A rigid substitution group reduces the FAs of the ARs.
2. The substitution of electron-withdrawing groups, such as F and CF_3 can significantly increase the FAs of the ARs.
3. The electron-abundant atoms such as O can be used to reduce the FAs.

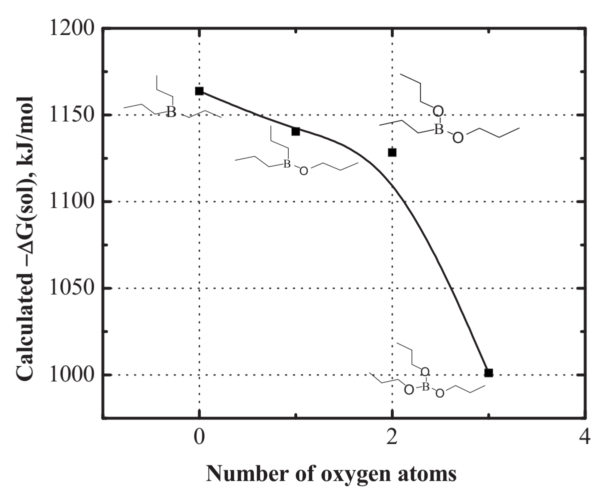

Figure 8. Impact of the number of the alkoxyl substitution groups on the calculated $-\Delta G(\mathrm{sol})$.

The calculations indicated that a promising AR can be specially formulated by deploying high fluorination to improve its chemical and electrochemical stability and by using the oxygen bridge to balance the FAs.

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